

Summary

PhD researcher in 3D vision, computer graphics, and generative modeling, with research and industry experience across ANU, KAUST, Tencent XR, and Baidu. My research explores how geometric structure, learned representations, and generative priors can be leveraged for efficient 3D visual modeling, with a current focus on articulated 3D object reconstruction and generation. My work has progressed from 3D point-cloud perception and generative human modeling to mesh-native generative learning and structured 3D object modeling.

- **Research Focus:** Generative modeling for 3D vision and graphics, geometry-aware learning, 3D reconstruction, mesh processing and articulated object modeling.
- **Experience:** Research across autonomous driving, XR, 3D human modeling, mesh-based generation, and articulated-object understanding, spanning both academic research and industrial R&D.
- **Expected graduation:** January 2028.
- **Internship availability:** Available year-round in 2027.

Research

End-to-End Articulated 3D Object Reconstruction

- Developed **Artic-O**, an end-to-end feed-forward framework for articulated 3D object reconstruction from sparse observations of two articulation states. Rather than optimizing each object independently at test time, Artic-O leverages a learned 3D geometry prior to infer complete shape representations and jointly predict movable-part segmentation and articulation parameters. This geometry-prior-driven formulation enables efficient structured 3D reasoning and reduces inference time to **0.32 seconds per object**, compared with approximately **9 minutes** for prior per-object optimization-based approaches.
- **Publication:** Xuyang Wang, Zhenyu Li, Jian Ding, Habib Slim, Peter Wonka, Hongdong Li, Mohammed Elhoseiny. “**Artic-O: End-to-End Articulated Object Reconstruction via Latent Geometry Learning**.” Conditionally accepted, ACM SIGGRAPH Asia 2026 Conference Papers.

Generative Learning on 3D Meshes with Manifold Diffusion

- Developed **DoubleDiffusion**, a mesh-native generative framework that incorporates intrinsic surface geometry directly into feature learning and denoising diffusion. By representing meshes as discrete manifolds and using heat diffusion derived from the Laplace–Beltrami operator for geometry-aware feature propagation, the model learns signal distributions directly on irregular 3D surfaces without UV parameterization. The resulting formulation scales to meshes with up to **100K vertices** while substantially improving computational efficiency over prior manifold-based approaches.
- **Publication:** Xuyang Wang, Ziang Cheng, Zhenyu Li, Jiayu Yang, Haorui Ji, Pan Ji, Mehrtash Harandi, Richard Hartley, Hongdong Li. “**DoubleDiffusion: Combining Heat Diffusion with Denoising Diffusion for Generative Learning on 3D Meshes**” Preprint; manuscript in preparation for 3DV 2027. [Project Page](#).

Monocular Depth Estimation with Adaptive Bins

- We study monocular depth estimation with a DETR-style decoder that progressively predicts adaptive depth bins and estimates depth maps in a coarse-to-fine manner.
- **Publication:** Zhenyu Li, Xuyang Wang, Junjun Jiang, Xianming Liu. “**BinsFormer: Revisiting Adaptive Bins for Monocular Depth Estimation**.” *IEEE Transactions on Image Processing*, 2024. [Project Page](#).

Research and Industry Experience

Visiting PhD Student

Vision-CAIR, KAUST, Saudi Arabia

Jan. 2026 – Jul. 2026

- Conducting research on articulated 3D object modeling, reconstruction, and generation for 3D vision and embodied applications, in collaboration with researchers at KAUST.
- Led the development and experimental evaluation of Artic-O, an end-to-end articulated object reconstruction framework based on learned 3D geometry priors.

Research Intern

Tencent XR Vision Lab, Canberra, Australia

Aug. 2022 – May. 2023

- Developed generative models for reconstructing full-body SMPL poses from sparse signals captured by a VR headset and hand controllers, exploring Transformer-, VAE/CVAE, and diffusion-based architectures.

Software Engineer Intern	Baidu Inc., Shanghai, China	Nov. 2021 – Jan. 2022
Developed a 3D point-cloud tracking pipeline for autonomous driving using nuScenes and internal datasets, including point-cloud visualization, data preprocessing, and neural object tracking based on CenterPoint.		
Research Intern	Corerain Technologies, Shenzhen, China	Dec. 2019 – Feb. 2020
Investigated hardware-level parallelism and acceleration opportunities for RNN, GRU, LSTM, and attention-based architectures.		

Education

PhD, Australian National University, Australia	Expected Jan. 2028
Tentative Thesis Topic: <i>Geometry-Aware and Efficient Generative Modeling for 3D Assets.</i> Supervisors: Prof. Hongdong Li, Prof. Mehrtaash Harandi and Prof. Richard Hartley.	
Master of Machine Learning and Computer Vision, Australian National University, Australia	2023
Thesis: <i>Generative Model Approaches for 3D Full-Body Human Pose Reconstruction from Sparse Joint Observations.</i>	
Bachelor of Electrical Engineering, University of Sydney, Australia	2020
Thesis: <i>Wireless Signal Modulation Classification with Deep Learning Methods.</i>	

Academic Service

Academic Tutor	Australian National University	2022 – 2024
COMP8610 Computer Graphics, Semester 1, 2024. Supervisor: Prof. Hongdong Li. COMP8536 Advanced Topics in Deep Learning in Computer Vision, Semester 2, 2022. Supervisor: Prof. Nick Barnes.		
Conference Reviewer: ECCV 2024, ECCV Workshops 2024, NeurIPS 2024, CVPR 2025, ICML 2025.		

Awards and Scholarships

University Research Scholarship, Australian National University	Jan. 2024 – Jan. 2028
Scholarship for international higher-degree research candidates, associated with the ANU HDR Fee Merit Scholarship covering full PhD tuition fees.	
HEX Singapore Program Scholarship	Jun. 2023 – Jul. 2023
Awarded by the Australian National University to support participation in the HEX Singapore program, including return airfare, accommodation, and living allowance.	
Vice-Chancellor’s Global Mobility Scholarship, University of Sydney	Jul. 2019 – Aug. 2019
Supported participation in the summer program <i>Global Cities as Centers of Knowledge Production</i> at Humboldt University of Berlin.	